

Direct Three-Block ADMM Can Cycle with an Identity Slack Block*

Author Name[†]

July 18, 2026

Abstract

Adding a nonnegative slack variable converts a two-block inequality problem into a three-block equality formulation with an identity third block and an explicit projection. We show that direct three-block ADMM may nevertheless fail even when the first two blocks are strongly convex quadratics. For $A = B = I_2$ and $\beta = 1$, we construct an explicit rational instance with a unique KKT point and a bounded non-KKT orbit of minimal period 66. An exact reduction to a four-dimensional signed piecewise-affine map yields a rational certificate that verifies the projection pattern and the original ADMM updates. The certified cycle persists under small perturbations, so the failure is not isolated. We also show that modifying only the multiplier step makes the former periodic initialization converge for every step size in a nondegenerate rational interval around $1/2$. Thus an identity slack block alone does not guarantee global convergence. The instance was discovered in an AI-assisted workflow using OpenAI's GPT-5.6 Sol model.

Keywords: alternating direction method of multipliers, multi-block optimization, slack variable, periodic orbit, piecewise-affine system, multiplier relaxation, rational certificate.

1 The question and the counterexample

The alternating direction method of multipliers (ADMM) is a standard method for structured convex optimization. Its classical convergence theory is strongest for two blocks, where the method is closely connected with Douglas–Rachford splitting and the proximal point algorithm [2, 5]. A direct Gauss–Seidel extension to three or more blocks is natural and easy to implement, but it is not convergent without additional assumptions in general [4].

A particularly tempting three-block structure arises when a linear inequality is converted into an equality by introducing a nonnegative slack variable. Starting from

$$\min_{x,y} \theta_1(x) + \theta_2(y) \quad \text{subject to} \quad Ax + By \leq b, \quad (1.1)$$

one obtains

$$\min_{x,y,z} \theta_1(x) + \theta_2(y) + \iota_{\mathbb{R}_+^m}(z) \quad \text{subject to} \quad Ax + By + z = b. \quad (1.2)$$

The third coefficient matrix is I_m , and the z -subproblem is the explicit orthogonal projection onto $\mathbb{R}_+^m$. These two properties are absent from a generic three-block model and suggest a concrete question:

*The executable certificate package is available at <https://github.com/ConanXu-math/identity-slack-admm-cycle-certificate>.

[†]Department of Mathematics, University Name. E-mail address: author@university.edu.

Does the identity coefficient and explicit orthant projection make the unmodified slack-last three-block ADMM converge from every finite initialization?

To state the iteration precisely, let $F = \theta_1 + \iota_X$ and $G = \theta_2 + \iota_Y$ be proper closed convex functions. We use the augmented Lagrangian convention

$$\mathcal{L}_\beta(x, y, z, \lambda) = F(x) + G(y) + \iota_{\mathbb{R}_+^m}(z) - \lambda^\top r + \frac{\beta}{2} \|r\|^2, \quad r = Ax + By + z - b. \quad (1.3)$$

The direct three-block ADMM performs

$$x^{k+1} \in \arg \min_x \mathcal{L}_\beta(x, y^k, z^k, \lambda^k), \quad (1.4a)$$

$$y^{k+1} \in \arg \min_y \mathcal{L}_\beta(x^{k+1}, y, z^k, \lambda^k), \quad (1.4b)$$

$$z^{k+1} \in \arg \min_{z \geq 0} \mathcal{L}_\beta(x^{k+1}, y^{k+1}, z, \lambda^k), \quad (1.4c)$$

$$\lambda^{k+1} = \lambda^k - \beta r^{k+1}. \quad (1.4d)$$

The order $x \rightarrow y \rightarrow z \rightarrow \lambda$ is part of the algorithm. Reordering the slack block or inserting a correction changes the iteration map.

Under the sign convention in (1.3), a KKT point $(x^*, y^*, z^*, \lambda^*)$ satisfies

$$A^\top \lambda^* \in \partial F(x^*) \quad B^\top \lambda^* \in \partial G(y^*) \quad (1.5a)$$

$$Ax^* + By^* + z^* = b \quad \lambda^* \in N_{\mathbb{R}_+^m}(z^*) \quad (1.5b)$$

Thus $z^* \geq 0$, $\lambda^* \leq 0$, and $(z^*)^\top \lambda^* = 0$. Readers using a nonnegative inequality multiplier may set $\mu = -\lambda$.

Known general multi-block counterexamples do not settle this identity-slack subclass, while existing positive results impose hypotheses not met by the orthant-indicator last block. Section 2 gives the precise boundary.

This paper gives a negative answer by constructing an exact periodic orbit inside the identity-slack subclass. We consider

$$\min_{x, y, z \in \mathbb{R}^2} \frac{1}{2} x^\top Q_1 x + \frac{1}{2} y^\top Q_2 y, \quad x + y + z = \bar{b}, \quad z \geq 0, \quad (1.6)$$

with rational $Q_1, Q_2 \succ 0$ and rational $\bar{b}$. The problem is strictly convex in (x, y) and has a unique KKT point. Nevertheless, direct three-block ADMM has a non-KKT periodic orbit that never touches a projection boundary and has mask word

$$\mathcal{W} = (00, 00, \underbrace{01, \dots, 01}_{64 \text{ times}}). \quad (1.7)$$

For $\beta = 1$, the signed projection state is $q^k = z^k + \lambda^k$. Thus 00 denotes $q_1^k < 0, q_2^k < 0$, whereas 01 denotes $q_1^k < 0, q_2^k > 0$.

Theorem 1.1 (Main result). *For the explicit rational data in Section 4 and the rational initialization defined by the period equation in Section 6, the unmodified direct three-block ADMM (1.4), with $A = B = I_2$ and $\beta = 1$, has a bounded non-KKT orbit of minimal period 66 that never meets a projection boundary. The problem has a unique KKT point, the projection-mask word of the orbit is $(00)^2(01)^{64}$, and all 132 coordinatewise projection inequalities are strict with a common margin greater than 10^{-3} . Consequently, an identity slack block does not guarantee unconditional global convergence of direct three-block ADMM.*

Consequently, the answer to the question above is no.

The strict certificate gives more than an isolated arithmetic example. Because the period system is nonsingular and every projection inequality has positive margin, the same primitive word remains admissible after all sufficiently small perturbations of $(Q_1, Q_2, \bar{b})$. We prove this open-neighborhood persistence after the exact certificate. We then ask what happens if only the multiplier step is relaxed. For the same QP and the same periodic initialization, an exact interval certificate shows convergence for every step length in a nondegenerate rational neighborhood of $1/2$, while the unrelaxed value 1 gives the cycle in the main theorem.

Proof roadmap. The remainder of the paper reduces the raw iteration to a finite word certificate, specifies the rational instance, and verifies the period-66 orbit exactly. It then proves persistence of the strict word and analyzes multiplier relaxation by a common Lyapunov ellipsoid, a finite-prefix capture certificate, and an exact Schur boundary. The executable certificates are recorded in Appendices A and B.

AI usage. This counterexample was developed through an AI-assisted research workflow using Codex agents powered by OpenAI’s GPT-5.6 Sol model. The agents were used to derive reduced maps, implement and run structured active-set searches, write exact checkers, perform internal adversarial audits, and assist with manuscript preparation. Section 4 records the mathematical discovery route and separates floating-point proposal from exact-rational acceptance.

The AI-assisted search and internal reviews record discovery provenance; they are not theorem evidence or external independent review. The theorem rests on the explicit derivations below and the exact certificate in the public source package. The authors remain responsible for the mathematical claims, citations, code, and final text.

2 Relation to existing ADMM theory

The two-block ADMM is connected with Douglas–Rachford splitting and the proximal point algorithm [5]. A third Gauss–Seidel minimization need not preserve that structure. Chen, He, Ye, and Yuan showed that the general direct multi-block method can diverge, including after a strongly convex extension [4]. Their examples establish that strong convexity alone is insufficient in the general model, but they do not use the identity-orthant slack triple $[A, B, I]$ and do not provide a bounded, strictly admissible active-set cycle of the form constructed here.

The closest same-order formulation appears in the slack-last discussion of He, Xu, and Yuan [7]. There the direct three-block sweep motivates a prediction–correction framework because it has no unconditional guarantee; the direct predictor is not certified by a slack-specific exact finite cycle. Our result supplies such a cycle for the unmodified predictor itself.

Positive multi-block theorems have sharply different assumption boundaries. Cai, Han, and Yuan obtain convergence from ordered strong convexity together with rank and penalty restrictions [3]. Lin, Ma, and Zhang prove a global result when the third objective is smooth and strongly convex and its condition number obeys a stringent bound [11]. The orthant indicator in our last block is neither smooth nor strongly convex, so these theorems do not apply. Their earlier inequality formulation uses a slack-first order and additional small-penalty conditions [10]; both the block order and the parameter regime differ from the slack-last map here.

Algorithmic modifications form a separate boundary. Prediction–correction schemes [8, 7] add a correction after the Gauss–Seidel predictor rather than accepting the direct sweep unchanged. The slack-QP method of Lew et al. [9] instead uses a two-block splitting and eliminates the slack from

an auxiliary update. Their convergence guarantees therefore concern different iteration maps and are not contradicted by the cycle below.

For quadratic programs, fixed active sets produce affine recurrences [1, 6]. Those analyses explain why an affine branch is the correct local object, but a spectral calculation on one branch does not certify a switching orbit. Our word crosses cells, so affine period closure, strict active-set admissibility, and the bridge back to the original ADMM subproblems must all be proved together.

Against this background, the first distinctive feature is the construction itself: identity slack, slack-last order, pure strongly convex rational quadratics, and a strictly admissible bounded finite cycle occur together. To the best of our knowledge, this precise combination has not appeared in the cited ADMM literature. This is not a claim that general multi-block nonconvergence or strongly convex counterexamples are new.

A second distinctive feature is the proof architecture. The projection and multiplier are combined into one signed state, so the original ADMM becomes a finite-family piecewise-affine system. Solving an affine period equation is only a candidate generator: the proposed word is accepted only after the actual projection and the original subproblem equations have both been checked.

Figure 1 summarizes this logic.

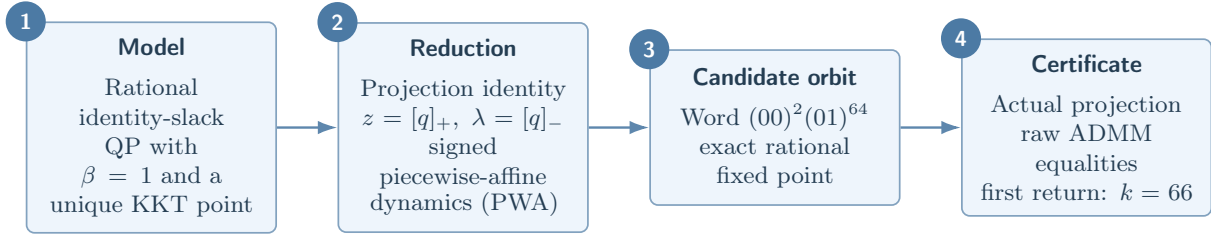


Figure 1: Proof architecture. The affine word supplies a candidate cycle; strict projection signs and the original subproblem equations turn it into a certificate for the unmodified three-block ADMM.

3 From the slack projection to a finite word certificate

This section turns the original ADMM iteration into a finite algebraic certificate. The identity slack first eliminates the complementary pair (z, λ) ; strict sign patterns then select affine branches of a four-dimensional recurrence.

Lemma 3.1 (Projection and complementarity at every step). *Define*

$$q^{k+1} = b - Ax^{k+1} - By^{k+1} + \lambda^k / \beta. \quad (3.1)$$

Then (1.4c)–(1.4d) are exactly

$$z^{k+1} = \Pi_{\mathbb{R}_+^m}(q^{k+1}), \quad \lambda^{k+1} = \beta(q^{k+1} - z^{k+1}). \quad (3.2)$$

Consequently,

$$z^{k+1} \geq 0, \quad \lambda^{k+1} \leq 0, \quad (z^{k+1})^\top \lambda^{k+1} = 0, \quad q^{k+1} = z^{k+1} + \lambda^{k+1} / \beta. \quad (3.3)$$

Proof. Completing the square in the z -subproblem gives

$$\min_{z \geq 0} \frac{\beta}{2} \|z - q^{k+1}\|^2,$$

which proves the projection formula. Substitution into the multiplier update gives $\lambda^{k+1} = \beta(q^{k+1} - z^{k+1})$. The coordinatewise identities $q = [q]_+ + [q]_-$, $z = [q]_+$, and $q - z = [q]_-$ give (3.3). $\square$

The lemma therefore makes the complementary slack–multiplier pair exact at every step, not merely at a limit. To eliminate the pair and expose the finite branch structure, specialize to (1.6) and $\beta = 1$. The raw iteration is

$$x^{\text{new}} = (Q_1 + I)^{-1}(\lambda - y - z + \bar{b}), \quad (3.4a)$$

$$y^{\text{new}} = (Q_2 + I)^{-1}(\lambda - x^{\text{new}} - z + \bar{b}), \quad (3.4b)$$

$$q^{\text{new}} = \bar{b} - x^{\text{new}} - y^{\text{new}} + \lambda, \quad (3.4c)$$

$$z^{\text{new}} = [q^{\text{new}}]_+, \quad (3.4d)$$

$$\lambda^{\text{new}} = [q^{\text{new}}]_-. \quad (3.4e)$$

Here positive and negative parts are **componentwise**, with $[a]_+ = \max\{a, 0\}$ and $[a]_- = \min\{a, 0\}$.

After one projection and multiplier update, Lemma 3.1 gives

$$q = z + \lambda, \quad z = [q]_+, \quad \lambda = [q]_-. \quad (3.5)$$

Let

$$M = (Q_1 + I)^{-1}, \quad N = (Q_2 + I)^{-1}, \quad g = (I - M)\bar{b}. \quad (3.6)$$

Proposition 3.2 (Exact signed-state recurrence). *On the state $s = (y, q) \in \mathbb{R}^4$, the raw iteration (3.4) is exactly equivalent to*

$$p = My - (I - M)|q| + g, \quad (3.7a)$$

$$y^{\text{new}} = Np, \quad (3.7b)$$

$$q^{\text{new}} = (I - N)p + [q]_+. \quad (3.7c)$$

Conversely, a signed-state orbit of (3.7) reconstructs an orbit of the original ADMM by $z = [q]_+$, $\lambda = [q]_-$, and

$$x^{\text{new}} = M(\bar{b} - y - |q|). \quad (3.8)$$

Proof. The identity $\lambda - z = [q]_- - [q]_+ = -|q|$ turns (3.4a) into (3.8). Substituting it into (3.4b) gives

$$\bar{b} - x^{\text{new}} - |q| = My - (I - M)|q| + (I - M)\bar{b} = p,$$

and hence $y^{\text{new}} = Np$. Finally,

$$q^{\text{new}} = \bar{b} - x^{\text{new}} - Np + [q]_- = p + |q| - Np + [q]_- = (I - N)p + [q]_+.$$

Every step is reversible once (3.5) is imposed, proving exact equivalence. $\square$

For a strict orthant mask $D = \text{diag}(\mathbf{1}_{\{q_i > 0\}})$ —the active-set indicator of the projection—set $S_D = 2D - I$. On that orthant, $|q| = S_D q$ and $[q]_+ = Dq$, so

$$s^{\text{new}} = T_D s + h, \quad (3.9)$$

where

$$T_D = \begin{pmatrix} NM & -N(I - M)S_D \\ (I - N)M & D - (I - N)(I - M)S_D \end{pmatrix}, \quad h = \begin{pmatrix} Ng \\ (I - N)g \end{pmatrix}. \quad (3.10)$$

This is an exact branch formula, not a derivative or local approximation.

We attach D_k to the *source* q^k . Thus D_k is used in the map from s^k to s^{k+1} , while the sign of q^{k+1} determines D_{k+1} . This convention is important: confusing source and target masks would shift every projection test by one phase.

Let $\mathcal{W} = (D_0, \dots, D_{L-1})$ be a proposed source-mask word and lift each affine branch to

$$L_D = \begin{pmatrix} T_D & h \\ 0 & 1 \end{pmatrix}. \quad (3.11)$$

Chronological composition gives

$$L_{\mathcal{W}} = L_{D_{L-1}} \cdots L_{D_0} = \begin{pmatrix} P_{\mathcal{W}} & a_{\mathcal{W}} \\ 0 & 1 \end{pmatrix}. \quad (3.12)$$

If $I - P_{\mathcal{W}}$ is nonsingular, the only possible periodic state for this word is

$$s^0 = (I - P_{\mathcal{W}})^{-1} a_{\mathcal{W}}. \quad (3.13)$$

Equation (3.13) proves closure only for the proposed affine composition. It represents a genuine ADMM orbit if and only if every state generated from s^0 lies in the strict orthant assigned by its source mask. Writing $d_{k,i} \in \{0, 1\}$ for the i -th diagonal entry of D_k , define the signed itinerary margin

$$\gamma(\mathcal{W}) = \min_{0 \leq k < L} \min_i (2d_{k,i} - 1)q_i^k. \quad (3.14)$$

Then $\gamma > 0$ certifies strict admissibility, $\gamma = 0$ signals a projection tie, and $\gamma < 0$ invalidates the proposed word. This separation between affine closure and projection admissibility is the central logical interface of the proof.

Proposition 3.3 (Strict word certificate). *Consider the quadratic model (1.6), for which the raw ADMM update is single valued and is equivalent to (3.7). Let $\mathcal{W} = (D_0, \dots, D_{L-1})$ be a word of strict source masks. Suppose there is a state $s^0 = (y^0, q^0)$ whose branch iterates satisfy*

$$s^{k+1} = T_{D_k} s^k + h \quad (0 \leq k < L), \quad s^L = s^0, \quad (3.15)$$

and whose signed itinerary margin $\gamma(\mathcal{W})$ is positive. Then $(s^k)_{k=0}^{L-1}$ is an actual periodic orbit of the signed recurrence. If, in addition, $\mathcal{W}$ is primitive—that is, $\mathcal{W} \neq \mathcal{V}^r$ for every shorter word $\mathcal{V}$ and integer $r \geq 2$ —then its minimal period is L .

With indices interpreted modulo L , the reconstruction

$$z^k = [q^k]_+, \quad \lambda^k = [q^k]_-, \quad x^k = M(\bar{b} - y^{k-1} - |q^{k-1}|) \quad (3.16)$$

is an actual raw-ADMM orbit with the same minimal period.

Proof. Positive margin makes the strict source mask unique:

$$D_k = \text{diag}(\mathbf{1}_{\{q_i^k > 0\}}).$$

Consequently, every affine branch in (3.15) is exactly the branch selected by the componentwise positive part in (3.7); closure therefore proves actual periodicity.

Let p be the minimal positive return time. Determinism and $s^L = s^0$ imply $p \mid L$: writing $L = ap + r$, $0 \leq r < p$, gives $s^L = s^r = s^0$, hence $r = 0$. Strictness also gives $D_{k+p} = D_k$, so $\mathcal{W} = (D_0, \dots, D_{p-1})^{L/p}$. Primitivity rules out $p < L$. Proposition 3.2, applied cyclically with $s^{-1} = s^{L-1}$, proves the raw reconstruction. Conversely, a return of the raw state would return $s^k = (y^k, z^k + \lambda^k)$, so the raw orbit cannot have a shorter period. $\square$

Remark 3.4 (Role of the period system). Nonsingularity of $I - P_{\mathcal{W}}$ is a convenient way to compute a unique candidate state through (3.13); it is not a soundness assumption of Proposition 3.3. Closure and positive margin establish a genuine orbit, while primitivity is the separate minimal-period obligation.

4 Construction of the rational instance

Because the signed recurrence depends on the resolvents $M = (Q_1 + I)^{-1}$ and $N = (Q_2 + I)^{-1}$, it is clearer to parameterize those matrices directly rather than begin with two dense Hessians. We use five short rational inputs,

$$\varepsilon = \frac{1}{1000}, \quad \mu = \frac{8957}{10000}, \quad \nu = \frac{999}{1000}, \quad \omega_M = (-1, 20)^\top, \quad \omega_N = (-1, 10)^\top. \quad (4.1)$$

Define

$$M = \varepsilon I + (\mu - \varepsilon) \frac{\omega_M \omega_M^\top}{\omega_M^\top \omega_M}, \quad N = \varepsilon I + (\nu - \varepsilon) \frac{\omega_N \omega_N^\top}{\omega_N^\top \omega_N}, \quad (4.2)$$

$$Q_1 = M^{-1} - I, \quad Q_2 = N^{-1} - I. \quad (4.3)$$

The values in (4.1) were not obtained by searching the entries of two dense Hessians. The signed reduction makes M and N the natural design variables. Giving them a small common eigenvalue and two different rank-one directions automatically preserves $0 \prec M, N \prec I$, hence $Q_1, Q_2 \succ 0$, while retaining a genuinely two-dimensional noncommuting coupling. At the parent point $(\mu_0, \nu_0) = (0.9, 0.999)$, the 01 branch has a dominant complex pair whose argument is approximately 0.095 , close to $2\pi/66$. This suggests a long residence in the 01 cell, with a few 00 steps serving as a return segment; Figure 3 displays the same mechanism at the final rational parameters. We therefore searched the run-length family $\mathcal{W}_{a,b} = (00)^a(01)^b$ rather than unrestricted long words.

For each trial (a, b, μ, ν) , the initial state was eliminated rather than treated as an optimization variable: we composed the affine word map and, whenever its fixed system was nonsingular, solved analytically for its unique fixed state. The candidate was then scored by the smallest signed projection margin along the resulting orbit. A differential-evolution screen over the two slow eigenvalues located a positive-margin candidate for $(a, b) = (2, 64)$. This stage proposed the word and a parameter neighborhood only; floating-point closure and positive numerical margins were not used as theorem evidence.

After that discovery, the floating-point parameters and orbit were discarded. A finite nearby grid of short rationals produced the values μ and ν in (4.1); they are not truncations of a floating-point trajectory. Each resulting rational QP was rebuilt from scratch and submitted to the exact acceptance procedure of Section 6; the displayed pair is one that passed. Thus the search selects a candidate, while all theorem evidence comes from exact arithmetic. Proposition 7.1 later explains why the certified word persists locally, but that openness result plays no role in accepting the instance.

The right-hand side is fixed by a simple target KKT pair. Set

$$z^* = (0, 1)^\top, \quad \lambda^* = (-1, 0)^\top, \quad \bar{b} = Q_1^{-1} \lambda^* + Q_2^{-1} \lambda^* + z^*. \quad (4.4)$$

Thus $\bar{b}$ is determined by the Hessians rather than introduced as an additional search parameter.

Equivalently,

$$Q_1 = \frac{1}{3591757} \begin{pmatrix} 3579218243 & 178940000 \\ 178940000 & 9365243 \end{pmatrix}, \quad (4.5)$$

$$Q_2 = \frac{1}{100899} \begin{pmatrix} 99800101 & 9980000 \\ 9980000 & 998101 \end{pmatrix}, \quad (4.6)$$

and

$$\bar{b} = \begin{pmatrix} -418394646086/42200300457 \\ 470482042273/4688922273 \end{pmatrix}. \quad (4.7)$$

The rank-one structure makes the spectra transparent:

$$\sigma(M) = \left\{ \frac{1}{1000}, \frac{8957}{10000} \right\}, \quad \sigma(N) = \left\{ \frac{1}{1000}, \frac{999}{1000} \right\}. \quad (4.8)$$

Therefore

$$\sigma(Q_1) = \left\{ 999, \frac{1043}{8957} \right\}, \quad \sigma(Q_2) = \left\{ 999, \frac{1}{999} \right\}, \quad (4.9)$$

so $Q_1, Q_2 \succ 0$ and both ADMM quadratic subproblems have unique minimizers.

Proposition 4.1 (Unique KKT point). *The problem (1.6) with data (4.5)–(4.7) has the unique KKT point*

$$\begin{aligned} x^* &= \begin{pmatrix} -9365243/417824757 \\ 178940000/417824757 \end{pmatrix}, & z^* &= (0, 1)^\top, \\ y^* &= \begin{pmatrix} -998101/100899 \\ 9980000/100899 \end{pmatrix}, & \lambda^* &= (-1, 0)^\top. \end{aligned} \quad (4.10)$$

Proof. Direct rational multiplication gives

$$Q_1 x^* = \lambda^*, \quad Q_2 y^* = \lambda^*, \quad x^* + y^* + z^* = \bar{b}. \quad (4.11)$$

Moreover, $z^* \geq 0$, $\lambda^* \leq 0$, and $(z^*)^\top \lambda^* = 0$. These are precisely the KKT conditions for the sign convention in (1.3). Since $Q_1, Q_2 \succ 0$, the objective is strictly convex in (x, y) . The feasible set is convex, so the primal optimizer (x^*, y^*) is unique; the equality constraint then determines z^* uniquely. Moreover, every KKT point satisfies $\lambda = Q_1 x$ because $A = I_2$. Hence the primal optimizer determines the multiplier uniquely as $\lambda = Q_1 x^* = \lambda^*$. The displayed quadruple is therefore the unique optimizer–multiplier pair and the unique KKT point. $\square$

5 Geometry of the certified orbit

Before giving the exact certificate, we first display the orbit and the branch mechanism behind its long residence. In Figure 2(a), consecutive values of $q^k = z^k + \lambda^k$ form a closed loop. The two red squares are the 00 states, the remaining points are 01 states, and the black star is the unique KKT value $q^* = z^* + \lambda^* = (-1, 1)$. The orbit passes around this fixed point without visiting it. Panel (b) resolves the switching coordinate: $q_2^k < 0$ for phases 0, 1, becomes strictly positive at phase 2, and stays positive until the return.

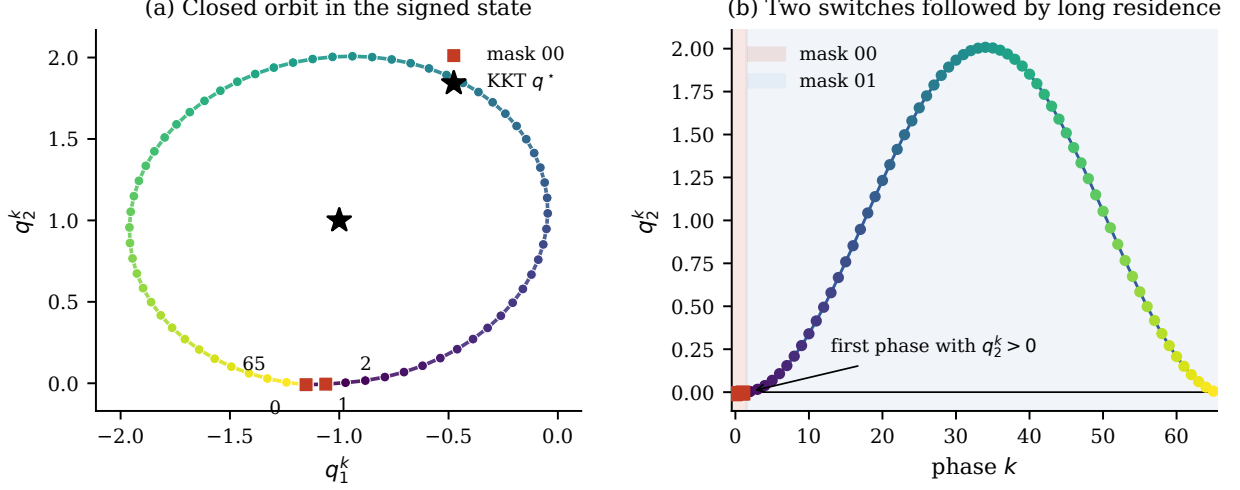


Figure 2: Decimal visualization of the exact rational cycle. Color records the phase within one period. The connected curve closes at phase 66, while the unique KKT point is not a cycle state.

The long 01 residence has a simple spectral explanation, although this explanation is not part of the exact proof. The 00 branch is contractive, with floating-point spectral radius $\rho(T_{00}) \approx 0.970582286560$. The 01 branch has the dominant pair

$$\lambda_{\pm}(T_{01}) \approx 0.9956552 \pm 0.0950692i, \quad |\lambda_{\pm}| \approx 1.000183721838.$$

Its angle $\theta \approx 0.095195477998$ differs from $2\pi/66$ by only -4.30×10^{-6} radians. Thus 64 consecutive 01 steps behave like a slightly expanding near-rotation, while the two 00 steps contract and reset the state into the long-residence cell. This gives a dynamical explanation for the run-length structure of $(00)^2(01)^{64}$; strict admissibility and closure still come solely from the rational certificate.

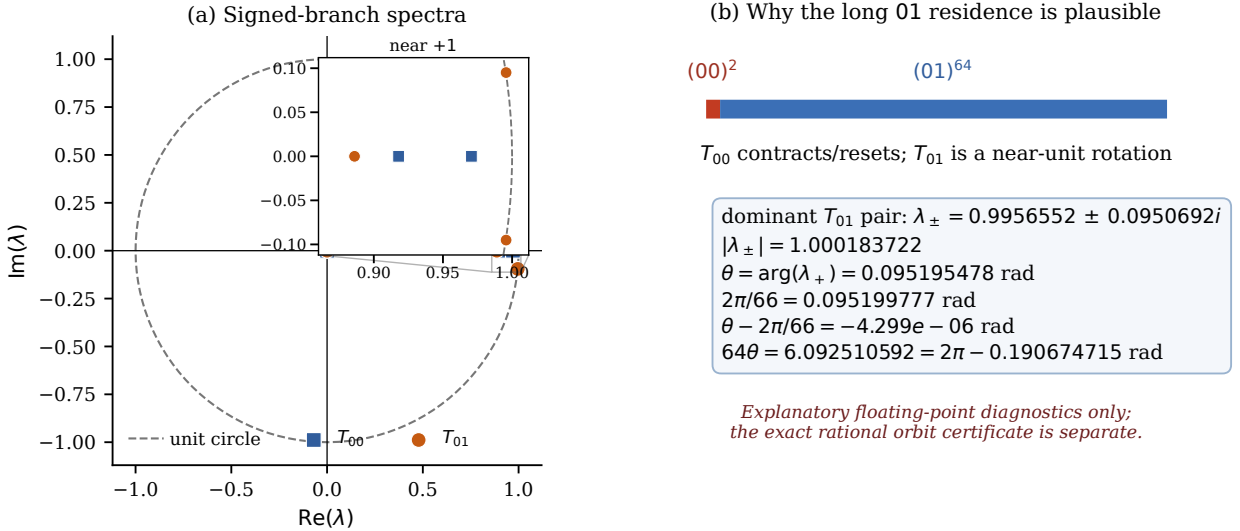


Figure 3: Floating-point mechanism diagnostic for the exact rational instance. The spectra explain why a long 01 residence is plausible, but they are not used to accept the word or prove period closure.

The geometry explains how a long switching cycle can arise; the exact closure and strict projection signs are established next.

6 The exact certificate and proof of the main theorem

For the candidate word $\mathcal{W} = (00)^2(01)^{64}$, only the masks 00 and 01 occur. Put

$$D_0 = \text{diag}(0, 0), \quad D_1 = \text{diag}(0, 1), \quad (6.1)$$

and let $L_0, L_1 \in \mathbb{Q}^{5 \times 5}$ be the affine lifts from (3.11). Because source masks act in chronological order from the right, the one-period map is

$$L_{\mathcal{W}} = L_1^{64} L_0^2 = \begin{pmatrix} P & a \\ 0 & 1 \end{pmatrix}. \quad (6.2)$$

Exact rational elimination gives $\det(I - P) \neq 0$. Define

$$s^0 = (I - P)^{-1}a \in \mathbb{Q}^4. \quad (6.3)$$

Then $s^{66} = Ps^0 + a = s^0$ exactly. Equation (6.3), not a rounded orbit, is the definition of the candidate initial state. For scale, its decimal approximation is

$$s^0 \approx (-9.9700010041, 99.6884664355, -1.1517671634, -0.0086191662)^\top, \quad (6.4)$$

where the first two coordinates are y^0 and the last two are q^0 . These are the coordinates of the pure-quadratic model (1.6).

Closure under (6.2) is not yet a proof about the original algorithm. Starting from (6.3), we therefore iterate (3.7) and compute $[q]_+$ from the exact sign of every coordinate rather than from the proposed selector. Exact rational arithmetic gives

$$q_1^k < 0 \quad (0 \leq k < 66), \quad (6.5a)$$

$$q_2^k < 0 \quad (k = 0, 1), \quad (6.5b)$$

$$q_2^k > 0 \quad (2 \leq k < 66). \quad (6.5c)$$

More strongly,

$$\min_{0 \leq k < 66} \min\{-q_1^k, (2d_k - 1)q_2^k\} > \frac{1}{1000}, \quad d_k = \begin{cases} 0, & k = 0, 1, \\ 1, & 2 \leq k < 66. \end{cases} \quad (6.6)$$

The actual minimum margin is approximately

$$0.00371052469443529102. \quad (6.7)$$

Thus the orbit never touches a projection boundary, its masks are uniquely determined, and the actual mask sequence is exactly $(00)^2(01)^{64}$. This proves strict admissibility rather than merely closure of a preset affine word.

For every phase set

$$z^k = [q^k]_+, \quad \lambda^k = [q^k]_-. \quad (6.8)$$

Using the source state (y^k, q^k) , reconstruct x^{k+1} by (3.8). Proposition 3.2 already shows algebraically that this reconstruction satisfies all four raw updates. The exact certificate additionally recomputes, at each of the 66 phases,

1. the x -subproblem stationarity equation;
2. the y -subproblem stationarity equation;
3. $q^{k+1} = \bar{b} - x^{k+1} - y^{k+1} + \lambda^k$;
4. the actual positive-part projection and multiplier update.

Every equality holds over $\mathbb{Q}$. To close the initialization explicitly, define the otherwise update-irrelevant starting x -component by

$$x^0 := M(\bar{b} - y^{65} - |q^{65}|). \quad (6.9)$$

Then the phase-65 update gives $x^{66} = x^0$, and the signed closure gives the other three components. Hence

$$(x^{66}, y^{66}, z^{66}, \lambda^{66}) = (x^0, y^0, z^0, \lambda^0). \quad (6.10)$$

Periodic extension yields the analogous equality after 66 steps from every phase.

The word $\mathcal{W}$ is primitive. Indeed, if it were $\mathcal{V}^r$ for some $r \geq 2$, then r would divide the total number of 00 symbols, so $r = 2$. The two length-33 copies would then contain the same number of 00 symbols, whereas the first contains both occurrences and the second contains none. Proposition 3.3 therefore proves that the signed and raw orbits have minimal period 66.

The initial state differs from the unique KKT point in Proposition 4.1; a nonconstant finite cycle cannot converge. The orbit is finite and hence bounded. This proves every assertion of Theorem 1.1.

The source package contains two separately implemented exact-rational checkers, one for the signed recurrence and one for the unreduced raw ADMM updates. Appendix A records the executable certificate.

7 Persistence of the strict cycle

The rational data make the counterexample exactly reproducible, but rationality is not what creates the cycle. The two strict ingredients in the certificate—a nonsingular period system and positive distance from every projection boundary—are open conditions.

Proposition 7.1 (Open family of period-66 counterexamples). *Let $\theta_0 = (Q_1, Q_2, \bar{b})$ denote the data in Section 4, with $A = B = I_2$ and $\beta = 1$. There is an open neighborhood*

$$\mathcal{U} \subset \mathbb{S}_{++}^2 \times \mathbb{S}_{++}^2 \times \mathbb{R}^2 \quad \text{of } \theta_0$$

such that every $\theta \in \mathcal{U}$ admits a strictly admissible, non-KKT orbit of minimal period 66 with word $(00)^2(01)^{64}$. Its initial signed state depends real-analytically on θ . In particular, $\mathcal{U}$ contains infinitely many rational instances whose corresponding periodic states are rational.

Proof. For the fixed word $\mathcal{W} = (00)^2(01)^{64}$, let

$$L_{\mathcal{W}}(\theta) = \begin{pmatrix} P_{\mathcal{W}}(\theta) & a_{\mathcal{W}}(\theta) \\ 0 & 1 \end{pmatrix}$$

be the affine period map obtained from the signed branches. On the positive definite cone, the resolvents $M = (Q_1 + I)^{-1}$ and $N = (Q_2 + I)^{-1}$ depend real-analytically on (Q_1, Q_2) ; hence so do $P_{\mathcal{W}}$ and $a_{\mathcal{W}}$. At θ_0 , the exact certificate gives $\det(I - P_{\mathcal{W}}(\theta_0)) \neq 0$. This remains true after shrinking to a neighborhood $\mathcal{U}_1$, and there the unique word-fixed state is

$$s^0(\theta) = (I - P_{\mathcal{W}}(\theta))^{-1} a_{\mathcal{W}}(\theta). \quad (7.1)$$

It is real-analytic in θ , as are the finitely many phase states generated from it.

Every signed branch margin is therefore continuous in θ . At θ_0 , their common minimum is greater than 10^{-3} by (6.6). Shrinking to a neighborhood $\mathcal{U} \subset \mathcal{U}_1$ preserves all 132 strict signs, so the actual positive-part projection realizes the same word. The word is primitive, and Proposition 3.3 then gives minimal period 66 for both the signed and raw ADMM states. The orbit is nonconstant and hence cannot contain a KKT fixed point.

Finally, rational positive definite matrices and rational vectors are dense in the displayed parameter space. For rational $\theta \in \mathcal{U}$, all branch data in (7.1) are rational, so the periodic state is rational as well. $\square$

Proposition 7.1 is qualitative: it proves that the failure occupies an open set but does not give the largest perturbation radius. A quantitative radius can in principle be obtained by combining a lower bound for $\sigma_{\min}(I - P_W)$ with Lipschitz bounds for the phase margins.

8 Stabilization by multiplier relaxation

The main theorem concerns the standard multiplier step. We now keep the same three primal minimizations, the same rational QP, and $\beta = 1$, but replace only the final update by

$$\lambda^{k+1} = \lambda^k - \tau(x^{k+1} + y^{k+1} + z^{k+1} - \bar{b}), \quad \tau > 0. \quad (8.1)$$

Thus $\tau = 1$ is the cycling method already proved above. When $\tau \neq 1$, the identity $\lambda = [q]_-$ need not persist, so the four-dimensional signed reduction is no longer available. Instead use the full essential state

$$w = (y, z, \lambda) \in \mathbb{R}^6.$$

Put

$$\begin{aligned} C_y &= (I - N)M, & C_z &= N + (I - N)M, & C_\lambda &= (I - N)(I - M), \\ C &= (C_y, C_z, C_\lambda), & d &= C_\lambda \bar{b}. \end{aligned} \quad (8.2)$$

Eliminating x^+ from the original subproblem equations gives $q^+ = Cw + d$. If the actual projection has selector $D = \text{diag}(\mathbf{1}_{\{q_i^+ > 0\}})$, then $z^+ = Dq^+$, and (8.1) becomes

$$\lambda^+ = (1 - \tau)\lambda + \tau(I - D)q^+.$$

Consequently the exact affine branch map is

$$w^+ = T_D(\tau)w + a_D(\tau), \quad (8.3)$$

where

$$T_D(\tau) = \begin{pmatrix} NM & -N(I - M) & N(I - M) \\ DC_y & DC_z & DC_\lambda \\ \tau(I - D)C_y & \tau(I - D)C_z & (1 - \tau)I + \tau(I - D)C_\lambda \end{pmatrix} = A_D + \tau E_D \quad (8.4)$$

and

$$a_D(\tau) = \begin{pmatrix} N(I - M)\bar{b} \\ Dd \\ \tau(I - D)d \end{pmatrix} = c_D + \tau f_D. \quad (8.5)$$

Theorem 8.1 (A common strict-branch Lyapunov interval). *Let w^* be a KKT state, suppose $q^* = Cw^* + d$ has no zero coordinate, and let $D_\star$ be its sign selector. If one matrix $H \succ 0$ satisfies*

$$F(\tau_\pm) := H - T_{D_\star}(\tau_\pm)^\top H T_{D_\star}(\tau_\pm) \succ 0, \quad (8.6)$$

then the same projection-safe ellipsoid is forward invariant for every $\tau \in [\tau_-, \tau_+]$, and the true projected iteration converges locally and linearly to w^ throughout that interval.*

Proof. Write $T_{D_\star}(\tau) = A + \tau E$. If $\tau = \eta\tau_- + (1 - \eta)\tau_+$, direct expansion gives

$$F(\tau) - \eta F(\tau_-) - (1 - \eta)F(\tau_+) = \eta(1 - \eta)(\tau_+ - \tau_-)^2 E^\top H E \succeq 0. \quad (8.7)$$

Thus endpoint positivity implies $F(\tau) \succ 0$ on the whole interval.

Let c_i be row i of C , omitting a zero row from the minimum, and define

$$\alpha = \min_i \frac{|q_i^*|^2}{c_i H^{-1} c_i^\top} > 0. \quad (8.8)$$

For $e = w - w^*$, the inequality

$$|c_i e|^2 \leq (c_i H^{-1} c_i^\top) e^\top H e$$

shows that $e^\top H e < \alpha$ preserves every strict sign of $q^* + C e$. The true projection therefore remains on $D_\star$. On that branch,

$$e^{+\top} H e^+ = e^\top H e - e^\top F(\tau) e,$$

so the ellipsoid is forward invariant and the error contracts linearly. $\square$

For the present instance, $q^* = (-1, 1)^\top$ and $D_\star = \text{diag}(0, 1)$. At $\tau = 1/2$, let $H \in \mathbb{Q}^{6 \times 6}$ be the exact solution of

$$H - T_{01}(1/2)^\top H T_{01}(1/2) = I_6. \quad (8.9)$$

Exact Sylvester tests give

$$H \succ 0, \quad F(49/100) \succ 0, \quad F(51/100) \succ 0. \quad (8.10)$$

Hence the same KKT-centered ellipsoid is a local basin for every $49/100 \leq \tau \leq 51/100$. The exact certificate gives the uniform contraction-factor upper bound 0.99985082128383873917.

The local result does not by itself say what happens to the period-66 initialization, which lies outside this ellipsoid. The next result connects that specific initial point to the local basin by a uniformly strict finite prefix.

Theorem 8.2 (Capture of the former periodic initialization). *For the rational QP and the period-66 initialization used in Theorem 1.1, the true projected iteration with (8.1) converges to the unique KKT point for every*

$$\frac{4999999999}{10000000000} \leq \tau \leq \frac{5000000001}{10000000000}. \quad (8.11)$$

Proof. Let w_c^k be the exact trajectory at $\tau = 1/2$, set $h = 10^{-10}$, and propagate a componentwise rational radius r_k by

$$r_{k+1} = |T_{D_k}(1/2)| r_k + h(|E_{D_k}| |w_c^k| + |E_{D_k}| r_k + |f_{D_k}|), \quad (8.12)$$

where absolute values are entrywise and $r_0 = 0$. Induction in (8.3) gives $|w^k(\tau) - w_c^k| \leq r_k$ for every τ in (8.11), provided the branch signs are retained. At each step, the exact bound

$$|q^{k+1}(\tau) - q_c^{k+1}| \leq |C| r_k$$

closes that induction: the first two selectors are 00, all subsequent selectors through step 232 are 01, and the smallest uniform signed margin is greater than 0.00075690253479337201.

Let $e_c = w_c^{232} - w^*$. The same componentwise enclosure gives

$$\sup_{\tau} \frac{(w^{232}(\tau) - w^*)^\top H(w^{232}(\tau) - w^*)}{\alpha} \leq 0.97537998816016382196 < 1. \quad (8.13)$$

Thus every trajectory in the interval enters the common invariant ellipsoid of Theorem 8.1 after the certified prefix and then converges linearly to w^* . $\square$

The fixed KKT branch has a larger exact local stability range than the common ellipsoid used above.

Proposition 8.3 (Exact local Schur boundary). *For $0 < \tau < 1$, the strict 01 branch of the present instance is Schur stable exactly when*

$$0 < \tau < \tau_c, \quad 0.9366061114 < \tau_c < 0.9366061115, \quad (8.14)$$

where τ_c is the unique root in $(0, 1)$ of the integer cubic G recorded in Appendix B.

Proof. Exact elimination factors the characteristic polynomial as

$$\det(zI - T_{01}(\tau)) = \frac{z(z + \tau - 1)}{40501000000000000} Q_\tau(z), \quad (8.15)$$

with Q_τ quartic. A real Schur recursion reduces stability of the quartic to finitely many univariate sign conditions. Exact factorization shows that every factor is positive on $(0, 1)$ except the single cubic $G(\tau)$. A Sturm sequence proves that G has exactly one root there, and rational evaluation gives the strict bracket in (8.14). The factors z and $z + \tau - 1$ are Schur stable throughout $0 < \tau < 1$, completing the equivalence. $\square$

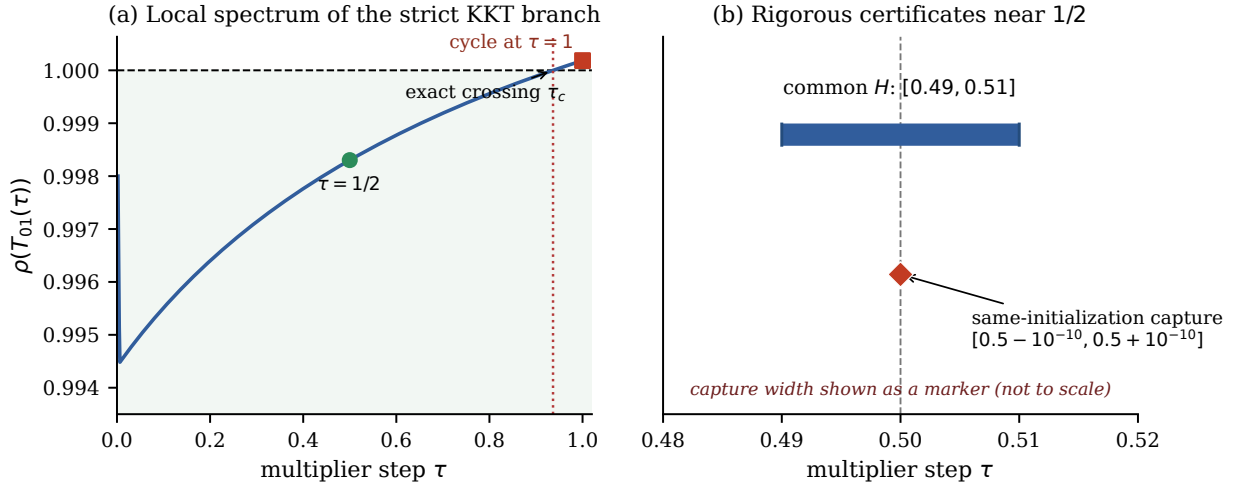


Figure 4: What multiplier relaxation changes for the fixed rational instance. Panel (a) is a decimal visualization of the exact 01-branch spectrum; the crossing is certified by the rational bracket for τ_c , not by the plot. Panel (b) shows the rigorously certified common-Lyapunov and same-initialization capture intervals near $1/2$; the capture marker is not drawn to its 2×10^{-10} width.

The three statements have deliberately different scopes. The Schur boundary is local to one strict KKT branch, the common ellipsoid is a uniform local certificate, and Theorem 8.2 concerns one fixed initial state. None is a global convergence theorem for arbitrary initializations or for the full identity-slack model class.

9 Consequences, limitations, and open questions

The counterexample settles only the unconditional claim. It gives bounded periodic nonconvergence, not unbounded divergence, and period 66 is the minimal period of this orbit rather than a lower bound for the entire identity-slack class. Proposition 7.1 does show that the phenomenon is structurally persistent: exact rationality selects a convenient certificate, not an isolated dynamical accident. Conditional convergence results based on block order, additional smoothness, penalty restrictions, proximal terms, or correction steps concern different regimes and remain unaffected.

The identity projection is nevertheless useful: it enforces complementarity at every standard step and exposes a finite piecewise-affine state space. What fails is the inference from those facts to global contraction. In the constructed instance, a near-unit rotation on the 01 branch and two resetting 00 steps balance to form a closed switching orbit. Multiplier relaxation breaks this balance. Theorems 8.1 and 8.2 prove one local stabilization mechanism and one finite-prefix route into it, without promoting either statement to global convergence.

Several quantitative questions remain. What is the largest explicit parameter neighborhood on which the period-66 word persists? What is the maximal multiplier-step interval for which the former periodic initialization converges, and what bifurcation separates it from the cycle at $\tau = 1$? Can one find a shorter strict rational cycle in the same subclass, or sharp conditions on Q_1, Q_2, β that exclude the long-residence mechanism? How does moving the slack block alter the admissible word structure? These questions seek the boundary between safe and unsafe identity-slack instances, rather than an unconditional theorem already ruled out by Theorem 1.1.

References

- [1] Daniel Boley, *Local linear convergence of the alternating direction method of multipliers on quadratic or linear programs*, SIAM Journal on Optimization **23** (2013), no. 4, 2183–2207.
- [2] Stephen Boyd, Neal Parikh, Eric Chu, Borja Peleato, and Jonathan Eckstein, *Distributed optimization and statistical learning via the alternating direction method of multipliers*, Foundations and Trends in Machine Learning **3** (2011), no. 1, 1–122.
- [3] Xinyu Cai, Deren Han, and Xiaoming Yuan, *The direct extension of ADMM for three-block separable convex minimization models is convergent when one function is strongly convex*, Computational Optimization and Applications **66** (2017), 39–73.
- [4] Caihua Chen, Bingsheng He, Yinyu Ye, and Xiaoming Yuan, *The direct extension of ADMM for multi-block convex minimization problems is not necessarily convergent*, Mathematical Programming **155** (2016), 57–79.
- [5] Jonathan Eckstein and Dimitri P. Bertsekas, *On the Douglas–Rachford splitting method and the proximal point algorithm for maximal monotone operators*, Mathematical Programming **55** (1992), 293–318.

- [6] Euhanna Ghadimi, André Teixeira, Iman Shames, and Mikael Johansson, *Optimal parameter selection for the alternating direction method of multipliers (ADMM): Quadratic problems*, IEEE Transactions on Automatic Control **60** (2015), no. 3, 644–658.
- [7] Bingsheng He, Shengjie Xu, and Xiaoming Yuan, *Extensions of ADMM for separable convex optimization problems with linear equality or inequality constraints*, Handbook of Numerical Analysis, vol. 24, Elsevier, 2023, pp. 511–557.
- [8] Bingsheng He and Xiaoming Yuan, *A class of ADMM-based algorithms for three-block separable convex programming*, Computational Optimization and Applications **70** (2018), no. 3, 791–826.
- [9] Thomas Lew, Marcus Greiff, John Subosits, and Brian Plancher, *Solving quadratic programs with slack variables via ADMM without increasing the problem size*, arXiv:2511.08451, 2025, Revised 2026; European Control Conference (ECC) 2026.
- [10] Tianyi Lin, Shiqian Ma, and Shuzhong Zhang, *On the global linear convergence of the ADMM with multiblock variables*, SIAM Journal on Optimization **25** (2015), no. 3, 1478–1497.
- [11] ———, *Global convergence of unmodified 3-block ADMM for a class of convex minimization problems*, Journal of Scientific Computing **76** (2018), no. 1, 69–88.

A Executable certificate package

The executable certificate package is available in the GitHub repository <https://github.com/ConanXu-math/identity-slack-admm-cycle-certificate>. The executable certificate is distributed as source files rather than as a multi-page typeset code listing. This keeps the mathematical article readable while preserving the complete machine-checkable object in the submission package. Its components are:

- `python/strict_cycle_certificate.py`, the six-dimensional unreduced essential-state (y, z, λ) checker;
- `python/signed_cycle_certificate.py`, the separately implemented four-dimensional (y, q) checker;
- Run `python/verify_certificate_pair.py` to regenerate both certificates. The same driver writes `certificates/instance_manifest.json`;
- `certificates/certificate_raw.json` and `certificates/certificate_signed.json`, the stable machine-readable outputs.
- `certificates/orbit_66.json`, the complete exact and decimal rendering of all 66 cyclic phases. It is a deterministic data export of the accepted certificate, not a separate checker.

The two checkers use separate state representations and do not import one another’s recurrence. Their exact agreement is an internal software cross-check, not external independent review. The manifest freezes the five rational design parameters, the word $(00)^2(01)^{64}$, the pure-quadratic formulation, and canonical hashes of the instance, initial state, orbit, checker sources, and JSON outputs.

The proposed word enters only the affine period solve. Each acceptance pass then derives the mask from the actual componentwise positive part and uses exact rational comparisons for closure, strict signs, KKT conditions, raw subproblem equations, and the absence of an earlier return. Any failed predicate produces a nonzero exit status. From this directory, the complete certificate is regenerated with Python 3.13.5 and SymPy 1.13.3 by

`python python/verify_certificate_pair.py.`

Figure files may then be regenerated with `python paper/generate_figures.py`; neither decimal figure generation nor the TeX build participates in certificate acceptance.

B Multiplier-relaxation certificate

The multiplier-relaxation artifact uses the same exact rational problem and periodic initialization as Appendix A. Its certifier reconstructs the six-dimensional branch map directly from the four original ADMM updates, checks the Sylvester minors of H and both endpoint residuals, propagates the 232-step componentwise rational enclosure, and applies exact Schur recursion and Sturm root counting. The release files are:

- `python/certify_relaxed_multiplier_half_convergence.py`, the exact $\tau = 1/2$ Lyapunov and true-prefix base checker;
- `python/certify_relaxed_multiplier_interval_theory.py`, the exact certifier;
- `certificates/relaxed_multiplier_certificate.json`, the machine-readable result containing rational predicates and hashes;
- `python/tests/test_relaxed_multiplier_interval_theory.py`, the regression and direct-replay checks.

From the source-package directory, the complete relaxation certificate is regenerated by

`python python/certify_relaxed_multiplier_interval_theory.py.`

For completeness, the quartic in (8.15) is

$$\begin{aligned}
Q_\tau(z) = & 40501000000000000z^4 \\
& + (3915586057000000\tau - 1169276880943000000)z^3 \\
& + (1123984258642286000 - 4222461747714000\tau)z^2 \\
& + (725214346843457\tau - 359717740103975543)z \\
& + 362404689543(1 - \tau).
\end{aligned} \tag{B.1}$$

The unique nontrivial boundary factor in the real Schur recursion is

$$G(\tau) = g_3\tau^3 + g_2\tau^2 + g_1\tau + g_0, \tag{B.2}$$

where

$$\begin{aligned}
g_3 &= 111794210406295556649228900462157733493, \\
g_2 &= 23105776975281816108275814441284422085171521, \\
g_1 &= -244157339715898821440243649673959463071543521, \\
g_0 &= 208410060660460340386576638889814578828638507.
\end{aligned} \tag{B.3}$$

The certificate records the complete factor list, exact root counts, endpoint signs, rational interval data, and the hashes of the large rational matrices. As with the cycle checkers, this is an internal executable cross-check and not external independent review or proof-assistant formalization.